STEM Website

Engineering Progress Summary

Updated 20 October 2026

Scope: the provided repository only. Friend's separate laptop and future game HTML are out of scope.

1. Executive status

The repository has progressed from a loosely connected frontend/backend prototype to a substantially more reproducible platform foundation. Authentication boundaries, server-authoritative progression, idempotent submissions, tutor failure states, repository checks, shared adapters, game modularization, and level parity checks are now represented in the local codebase.

Area	Status	Evidence
Backend authority	Complete enough for current baseline	Supabase migrations 001-013; authoritative submit procedures and idempotent wrappers
Shared frontend platform	Complete	js/game-adapter.js plus PhysicsSim/adapter.js and RoboMaze/adapter.js
RoboMaze modularization	Complete locally	levels.js, simulation.js, ui.js, controller.js, adapter.js
Physics modularization	Complete locally	levels.js, simulation.js, ui.js, controller.js, adapter.js
Level parity	Complete locally	scripts/verify-level-parity.mjs passes for 30 levels per game
Browser tests	Not yet implemented	Requires npm/Playwright setup and a test environment
Release verification	Manual gates remain	Hosted Supabase, browser smoke tests, and clean-machine checks

The current working tree contains the modularization and parity changes shown below. They are not automatically committed or pushed by this report.

2. Work completed

Backend and platform stabilization

The backend work established server-owned progression and rewards, revoked legacy client-trusted access where required, added idempotent submission wrappers, and documented the Supabase deployment/verification process. Tutor handling now distinguishes success, rate limiting, temporary provider unavailability, invalid responses, and rule-based fallback. The known Gemini 429/503 limitation remains operationally visible rather than being treated as a silent success.

Repository reproducibility

README and phase-7 documentation were brought to the 001-013 migration baseline. Repository verification checks required files, migration numbering, secret-pattern safety, web-quality constraints, and stale migration documentation. GitHub Actions runs syntax checks, repository checks, Node tests, Deno Edge Function tests, required-file checks, and secret scanning.

Frontend maintainability

Both game pages now use explicit controller entrypoints instead of large inline controllers. Inline event attributes were removed from the game pages. Platform operations are routed through the shared adapter factory, while simulation and rendering remain game-specific. This preserves the current visual/gameplay intent while creating boundaries that can be tested and extended.

Canonical data parity

The Supabase seed was selected as canonical. Frontend Physics and RoboMaze level definitions were updated to match the server seed, and scripts/verify-level-parity.mjs now compares IDs, counts, titles, instructions, unlock ordering, and validation-relevant configuration. The parity check currently passes.

3. Verification results

The following checks were run successfully in the available Node runtime:

- node scripts/verify-level-parity.mjs
- node scripts/verify-repo.mjs
- node scripts/verify-web-quality.mjs
- node --test tests/*.test.mjs
- node --check PhysicsSim/controller.js
- node --check RoboMaze/controller.js
- git diff --check

The Node test suite passed 34/34 tests. The exact npm commands could not be executed in this Codex environment because npm is not installed on the current PATH; this is an environment limitation, not a reported code failure.

4. Current changed files

The current uncommitted implementation work is concentrated in the following areas:

• PhysicsSim/controller.js and RoboMaze/controller.js: extracted controllers and event wiring.

• PhysicsSim/index.html and RoboMaze/index.html: module entrypoints and removal of inline handlers/controllers.

• PhysicsSim/levels.js and RoboMaze/levels.js: frontend definitions synchronized with Supabase seed data.

• PhysicsSim/ui.js and RoboMaze/ui.js: UI callbacks routed through adapters and data attributes.

• scripts/verify-level-parity.mjs: required canonical-data drift check.

• scripts/verify-web-quality.mjs: checks for direct game-page platform calls and inline handlers.

• package.json: npm check now includes level parity verification.

No Supabase migration was added in this final frontend phase. No production database change is required for the parity update because the already-selected seed data was treated as canonical.

5. Remaining work before new games

The roadmap is not fully complete until the following release gates are satisfied:

• Run npm ci, npm run check, npm test, and the Deno Edge Function test on a machine with npm and Deno available.

• Run browser smoke tests for login, protected access, level loading, reset, step/run, collision/failure, success, retry, unlock, refresh, and tutor fallback.

• Add Playwright only after the clean npm environment is available; do not expose service-role or Gemini secrets to browser tests.

• Generate and commit package-lock.json, then change CI from npm install to npm ci.

• Review CSP tightening after confirming all required pages no longer depend on inline scripts; inline styles may still require a separate cleanup.

• Complete hosted Supabase verification and record the result in the phase-7 guide.

• Only then begin the two new game modules.

Recommended immediate action

Review the current diff, run the checks on your local machine, open both games through the local server, and manually exercise the critical journeys. If those pass, commit the modularization/parity changes as one focused commit and push the branch for CI.

6. Safe handoff commands

From the repository root on a machine with Node/npm installed:

```
npm ci
```

```
npm run check
```

```
npm test
```

```
git diff --check
```

```
git status --short
```

```
git add PhysicsSim RoboMaze scripts/verify-level-parity.mjs
scripts/verify-web-quality.mjs package.json
```

```
git commit -m "modularize games and verify level parity"
```

```
git push origin feature/platform-stabilization
```

Do not run the commit commands until you have reviewed the diff and verified both games locally.